



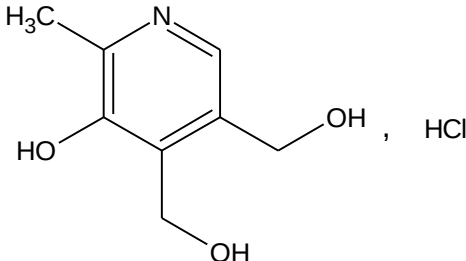
MATERIAL SAFETY DATA SHEET

Vitamin B₆ (Pyridoxine Hydrochloride)

1. Product Identification and Use

Product name	Pyridoxine Hydrochloride
Trade name	Pyridoxine Hydrochloride (Vitamin B ₆)
Company information	Manufacturer JIANGXI TIANXIN PHARMACEUTICAL CO.,LTD. Address: Le'anjiang Industrial Zone, Leping, Jiangxi, 333300, China Tel / Fax: +86-798-6702928
Use	Active pharmaceutical ingredient, nutritional ingredient.
Emergency telephone	+86-13958501873 Daniel Ting (丁文祥), Intl. Business Manager

2. Composition / Information on Ingredients

Chemical name	Pyridoxine hydrochloride (5-Hydroxy-6-methylpyridine-3,4-diyl)dimethanol hydrochloride
CAS No.	58-56-0
EINECS No.	200-386-2
Molecular formula	C ₈ H ₁₁ NO ₃ •HCl
Molecular weight	205.64 g/mol
Constitution	

3. Hazards Identification

Label precautionary statements	Irritating to respiratory system and skin. In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. Wear suitable protective clothing.
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4. First Aid Measures

Overexposure to	
Eyes	Immediately flush thoroughly with water for at least 15 minutes, occasionally lifting the upper and lower eyelids. Get medical aid immediately.
Skin	Flush skin with plenty of soap and water for at least 15 minutes while removing contaminated clothing and shoes. Get medical aid if irritation develops or persists. Wash clothing before reuse.
In case of Ingestion	Get medical aid immediately. Never give anything by mouth to an unconscious person. Do NOT induce vomiting. If conscious and alert, rinse mouth and drink 2-4 cupfuls of milk or water. Get medical aid if irritation or symptoms occur.
Inhalation	Remove from exposure to fresh air immediately. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Get medical aid if cough or other symptoms appear.
Notes to physician	Treat symptomatically and supportively.

5. Fire Fighting Measures

Suitable extinguishing media	Use water spray, dry chemical, carbon dioxide, or appropriate foam.
Special firefighting procedures	Wear self-contained breathing apparatus and protective clothing to prevent contact with skin and eyes.
Unusual fire and explosions hazards	Emits toxic fumes under fire conditions.

6. Accidental Release Measures

Spill response	Wear respirator, chemical safety goggles, rubber boots and heavy rubber gloves. Sweep up, place in a bag and hold for waste disposal. Avoid raising dust. Ventilate area and wash spill site after material pickup is complete.
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7. Handling and Storage

Handling	Wash thoroughly after handling, remove contaminated clothing and wash before reuse. Use with adequate ventilation. Avoid contact with eyes, skin, and clothing. Avoid ingestion and inhalation. Avoid prolonged, or repeated, skin contact.
Storage	Preserve in tight, light-resistant containers.

8. Exposure Controls / Personal Protection

Engineering controls	Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower. Use adequate ventilation to keep airborne concentrations low.
Personal protective equipment	
Respirators	Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Always use a NIOSH or European Standard EN 149 approved respirator when necessary.
Skin	Wear appropriate protective gloves to prevent skin exposure.
Eyes	Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

9. Physical and Chemical Properties

Color	White or almost white
Physical state	Crystalline powder or granular
Odor	Odorless
Bulk density	Powder: 0.48 ~ 0.60 g/cm ³ Granule: 0.80 ~ 0.88 g/cm ³
Tapped density	Powder: 0.92 ~ 1.00 g/cm ³ Granule: 0.90 ~ 0.95 g/cm ³
Solubility	Freely soluble in water (1:4.5), slightly soluble in alcohol (1:90) and acetone. Soluble in chloroform and insoluble in ether.
Melting point	205~209°C (with decomposition)

10. Stability and Reactivity

Chemical stability	Stable under normal temperatures and pressures.
Materials to avoid	Oxidants
Conditions to avoid	Incompatible materials, light, dust generation, excess heat, strong oxidants.
Hazardous decomposition products	Hydrogen chloride, nitrogen oxides, carbon monoxide, irritating and toxic fumes and gases, carbon dioxide, nitrogen.

11. Toxicological Information

Toxicity data	LD ₅₀ 4 g/kg (oral, rat) LD ₅₀ 5,500 mg/kg (oral, mouse)
Acute effects	May cause skin irritation.
Carcinogenicity	Not listed by ACGIH, IARC, NIOSH, NTP, or OSHA.

12. Ecological Information

Ecotoxicity

Toxicity to fish

OECD Guideline 203

Oncorhynchus mykiss/LC50 (96 h): >100 mg/L

The details of the toxic effect relate to the nominal concentration.

Aquatic invertebrates

OECD Guideline 202, part 1

Daphnia magna/EC50 (48 h): >100 mg/L

The details of the toxic effect relate to the nominal concentration.

Aquatic plants

OECD Guideline 201

Daphnia magna/EC50 (72 h): 5.3 mg/L

The details of the toxic effect relate to the nominal concentration.

Microorganisms / Effect on activated sludge

OECD Guideline 209 aerobic

activated sludge, domestic/EC20 (30 min): >1,000 mg/L

Assessment of aquatic toxicity

There is high probability that the product is not acutely harmful to aquatic organisms.

The inhibition of the degradation activity of activated sludge is not anticipated when introduced biological treatment plants in appropriate low concentrations.

Elimination Information

Test method

OECD 301E/92/69/EEC, C.4-B (aerobic), activated sludge, domestic

Method of analysis

DOC reduction

Degree of elimination

94% (28 d)

Evaluation

Biodegradable

13. Disposal Considerations

EPA Waste Number	Not available.
Treatment	Material does not have an EPA Waste Number and is not a listed waste, however consultation with a permitted waste disposal site (TSD) should be accomplished. Always contact a permitted waste disposal to assure compliance with all current local, state, and Federal Regulations. Dissolve or mix the material with a combustible solvent and burn in a chemical incinerator equipped with an afterburner and scrubber.
Waste from residues	Observe all national and local environmental regulations.

14. Transport Information

DOT Classification	Proper Shipping Name: CHEMICALS, N.O.S. RQ: Not applicable.
TDG Classification	Not applicable.
IMO/IMDG Classification	Proper Shipping Name: CHEMICALS, N.O.S. RQ: Not applicable.
ICAO/IATA Classification	Not applicable.

15. Regulatory Information

European Information

Irritability	R 36/37/38	Irritating to eyes, respiratory system and skin.
	S 26	In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
	S 36	Wear suitable protective clothing.

Remark This chemical substance is not classified in the Annex I of Directive 67/548/EEC.

U.S. Federal Regulations TSCA 8(b) inventory: VITAMIN B₆ HYDROCHLORIDE
 SARA 302/304/311/312 extremely hazardous substances: No products were found.
 SARA 302/304 emergency planning and notification: No products were found.
 SARA 302/304/311/312 hazardous chemicals: No products were found.
 SARA 311/312 MSDS distribution – chemical inventory – hazard identification: No products were found.
 SARA 313 toxic chemical notification and release reporting: No products were found.
 Clean Water Act (CWA) 307: No products were found.
 Clean Water Act (CWA) 311: No products were found.
 Clean air act (CAA) 112 accidental release prevention: No products were found.
 Clean air act (CAA) 112 regulated flammable substances: No products were found.
 Clean air act (CAA) 112 regulated toxic substances: No products were found.

State Regulations No products were found.
 California proposition 65: No products were found.

WHMIS (Canada) Not controlled under WHMIS (Canada).
 CEPA DSL: VITAMIN B₆ HYDROCHLORIDE
 This product has been classified in accordance with the hazard criteria of the Controlled Product Regulations and the MSDS contains all required information.

International Lists Australia (NICNAS): VITAMIN B₆ HYDROCHLORIDE
 Japan (MITI): VITAMIN B₆ HYDROCHLORIDE
 Korea (TCCL): VITAMIN B₆ HYDROCHLORIDE
 Philippines (RA6969): VITAMIN B₆ HYDROCHLORIDE

16. Other Information

Use

Active pharmaceutical ingredient, nutritional ingredient,
feed additive

Last update: March 26, 2012

The information in this MSDS is based on current scientific knowledge. It should not be taken as expressing or implying any warranty concerning product characteristics.